

Contracts for Gas Prioritization to Power Plants during Winter Emergencies

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Because our primary parameter on daily gas deliveries comes from S&P Global and is not publicly available, we are unable to share our final parameter files regarding gas consumption by sector (Residential, Commercial, Industrial customers, and Power Plants). However, several intermediate steps in our data process rely on publicly available information. All data sources used in this process are listed in the table below. For further details, please refer to the Appendix of this paper.

Parameter Source (Data provider/Form)	Purpose	Link
S&P Global (daily county-level gas deliveries data and spot natural gas price by pricing zone) (Not publicly available)	1. This source provides the daily county-level scheduled volumes of natural gas delivered at three Point Types (LDC, End User, and Power Plant) associated with 14 interstate pipelines serving 8 states in the Northeastern U.S. These three Point Types are then divided into two interconnection categories, LDC (LDC Point Type) and non-LDC (End User and Power Plant Point Types), for our further data process 2. This source also provides the daily spot natural gas price in four pricing zones (Transco Zone 6, Leidy, Algonquin, Iroquois)	• Operationally Available Capacity • Natural Gas Spot Prices
EIA Gas Consumption by End Use (monthly data) (Publicly available)	1. The gas deliveries data from <i>S&P Global</i> are not broken down by sector (Residential, Commercial, Industrial customers, and Power Plants). In contrast, the dataset here from EIA provides monthly sector-level gas deliveries data, which we can use to calculate the shares of gas delivered to each sector each month	• EIA Gas Consumption by End Use
EIA-176 Natural Gas Deliveries (annually data) (Publicly available)	1. The <i>EIA Gas Consumption by End Use</i> dataset provides monthly gas deliveries by sector, but it does not indicate whether the gas is delivered by an interstate pipeline (non-LDC) or by an LDC. In contrast, <i>EIA-176</i> provides annual sector-level gas deliveries data that includes this interconnection type distinction 2. We use the monthly shares by sector from <i>EIA Gas Consumption by End Use</i> to disaggregate the annual <i>EIA-176 Natural Gas Deliveries</i> into monthly volumes by sector and interconnection type 3. From these monthly <i>EIA-176</i> volumes, we calculate monthly shares by sector	• EIA-176 Gas Deliveries

	and interconnection type, and then apply those shares to the <i>S&P Global</i> data to split daily gas deliveries into different sectors	
EIA-176 Type of Operation and Sector Items (Publicly available)	1. This form allows us to determine whether a company reported in <i>EIA-176</i> as delivering gas to the four sectors is an interstate pipeline (non-LDC) or an LDC	• EIA-176 Operation Type
EIA-860 (data on power plants' status and characteristics) (Publicly available)	1. This form provides information on whether gas-fired power plants in the Northeastern U.S. are connected to an LDC or an interstate pipeline	• EIA-860 Power Plant Status
EIA-923 (monthly data on power plant operation and fuel use) (Publicly available)	1. This form provides monthly, state-level information on the types of contracts used to supply or transport amounts of natural gas to gas-fired power plants. Using this information, we can estimate state-level shares of firm and interruptible contracts held by power plants 2. These shares are then applied to <i>S&P Global</i> data to estimate the firm and interruptible contracts held by power plants	• EIA-923 Power Plant Fuel Use
EIA Manufacturing Energy Consumption Survey (MECS; Table 10.2 and Table 3.2) (Publicly available)	1. Every four years, this form reports annual data on fuel-switching capabilities for 21 manufacturing subsectors by region (Table 10.2) 2. This form also provides the total energy use by subsector and fuel type (Table 3.2) 3. Using the 2014 form data, we calculate two sets of shares of firm and interruptible contracts held by industrial customers: one representing a high firm share scenario (based on information from point 1) and the other representing a low firm share scenario (based on information from point 2)	• EIA MECS 2014
Bureau of Economic Analysis (BEA; annual data on county-level GDP) (Publicly available)	1. This source provides annual county-level data on manufacturing GDP. We use this information to calculate county-specific weights, which are then applied to the shares of firm and interruptible contracts held by industrial customers from the <i>MECS</i> data to obtain county-level weighted shares 2. These county-level weighted shares of firm and interruptible contracts are then applied to <i>S&P Global</i> data to estimate the firm and interruptible contracts held by industrial customers	• BEA County GDP
System data from Bent et al. (2018) (Publicly available)	Their GitHub provides data on the following: 1. Electric node location 2. Node connection for transmission line 3. Generator fuel type	• Bent Gas-grid System Data

	4. Fuel cost 5. Transmission line reactance 6. Transmission line limits 7. Generator-node connection 8. Power generation limits 9. Gas junction location 10. Gas injection limits 11. Pipeline length 12. Pipeline pressure 13. Compressor (we have additional steps to eliminate the compressors and determine whether a pipeline is active or passive) 14. Control valve (we have additional steps to eliminate the control valves)	• Bent, R., Blumsack, S., Van Hentenryck, P., Borraz-Sánchez, C., & Shahriari, M. (2018). Joint Electricity and Natural Gas Transmission Planning With Endogenous Market Feedbacks. <i>IEEE Transactions on Power Systems</i>, 33(6), 6397-6409.
EIA Average Tested Heat Rates by Prime Mover and Energy Source, 2013 – 2023 (Publicly available)	This source provides data on the heat rates of gas-fired generators, broken down by gas turbine and combined cycle	• EIA Heat Rate
FERC Form 714 (daily data) (Publicly available)	This source provides data on daily electric demand by balancing authority area and planning area	• FERC 714 data
Power system node information from Allen et al. (2008) (Publicly available)	Tables V and VI in this paper allow us to assign electric nodes to planning areas and then allocate the planning-area-level electric demand across individual nodes	• Allen, E. H., Lang, J. H., & Ilic, M. D. (2008). A Combined Equivalenced-Electric, Economic, and Market Representation of the Northeastern Power Coordinating Council U.S. Electric Power System. <i>IEEE Transactions on Power Systems</i>, 23(3), 896-907.